Scrum-Agile vs. Waterfall

By: Kristian Amick

Agile Roles and Why They Matter

Scrum combines clear responsibilities with teamwork and collaboration (West, n.d.; Schwaber & Sutherland, 2020).

Product Owner

- Represents customer and business needs, giving the team one clear voice on priorities.
- Organizes and prioritizes the product backlog.
- Helps the team understand which features provide the most value.

Scrum Master

- Helps the team follow Scrum effectively, keeping the process healthy and productive.
- Encourages communication and collaboration.
- Helps remove obstacles that slow progress.

Developers

- Build a usable product increment every sprint, turning the plan into working software.
- Plan and complete development work.
- Work together to maintain quality and meet the Definition of Done.

SNHU Travel Example

The Product Owner redirected the travel application toward wellness and detox travel. Developers adapted the Java slideshow images to fit the new focus, and the Scrum Master helped the team adapt to the change.

SDLC Phases in an Agile Approach

The phases of the software development life cycle (SDLC) can overlap and repeat during development instead of happening only once in a fixed sequence (Jackson et al., 2025).

- Planning: Set goals and prioritize user stories.
- Analysis: Understand customer needs and define acceptance criteria.
- Design: Plan the user experience and how the solution should work.
- Development: Write code and build the selected features.
- Testing: Check the software, find defects, and fix problems.
- Deployment: Release a usable product increment to users.
- Maintenance: Monitor the product, fix issues, and make improvements.

  Key Point: In the SNHU Travel project, the wellness change meant revisiting planning and design, updating the slideshow images in development, and checking the result in testing. Deployment and maintenance continue as the product is released and improved. A sprint should produce a usable increment that meets the team's Definition of Done (Schwaber & Sutherland, 2020).

How Waterfall Would Have Changed the Project

Waterfall follows a more sequential process, while Agile allows teams to adjust priorities and improve the product through short development cycles (Jackson et al., 2025; Radigan, n.d.).

Waterfall Approach

- The original travel application requirements and design would be approved.
- Developers would build the slideshow based on the approved plan.
- The change toward wellness and detox travel would require revisiting the requirements and design.
- The revised plan would need approval before the affected work continued.
- The slideshow would be updated and tested before release.

Scrum-Agile Approach

- The Product Owner would update the product backlog to reflect the wellness-travel focus.
- Developers would select the appropriate work for a sprint.
- The Java slideshow images would be updated to match the new direction.
- The team would check the changes and gather feedback.
- The Product Owner could adjust priorities for a future sprint.

Main Difference

With Waterfall, the change could require revising previously approved work and following a formal change process. With Agile, the team could adjust its priorities and update the slideshow in a future sprint. This example shows why Agile was a practical choice when the project's direction changed.

Choosing the Right Approach

The best development method depends on the requirements, risks, customer involvement, and delivery constraints (Radigan, n.d.).

Consider Waterfall When

- Requirements are stable and clearly defined.
- Approvals and documentation must be completed upfront.
- The project has predictable stages.
- Major changes are not expected.

Consider Agile When

- Customer needs may change.
- Frequent feedback is available.
- The product can be delivered in useful increments.
- The team can collaborate and test continuously.

Recommendation for SNHU Travel: Scrum-Agile

The shift toward wellness and detox travel showed why adaptability mattered. Agile allowed the team to revise the product direction instead of treating the original plan as unchangeable.

Success still depends on clear priorities, active Product Owner feedback, realistic sprint goals, and ongoing testing. Research on Scrum also highlights the importance of applying Scrum consistently and involving end users in the development process (Alami & Krancher, 2022).

References

Alami, A., & Krancher, O. (2022). How Scrum adds value to achieving software quality? *Empirical Software Engineering, 27*(7), Article 165. https://doi.org/10.1007/s10664-022-10208-4

Jackson, G., Holdsworth, J., & Kosinski, M. (2025). *What is the software development life cycle (SDLC)?* IBM. https://www.ibm.com/think/topics/sdlc

Radigan, D. (n.d.). *Waterfall vs. Agile: The differences in project management methodologies*. Atlassian. https://www.atlassian.com/agile/project-management/project-management-intro

Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide: The definitive guide to Scrum: The rules of the game*. https://scrumguides.org/scrum-guide.html

West, D. (n.d.). *Agile scrum roles and responsibilities*. Atlassian. https://www.atlassian.com/agile/scrum/roles